

Practice Projects

For practice, write a function for each of the following tasks:

1. Iteratively calculate the sum of the integer series from 1 to n . This is similar to the `factorial()` function, except it performs addition instead of multiplication. For example, `sumSeries(1)` returns 1, `sumSeries(2)` returns 3 (that is, $1 + 2$), `sumSeries(3)` returns 6 (that is, $1 + 2 + 3$), and so on. This function should use a loop instead of recursion. Take a look at the *factorialByIteration.py* program in this chapter for guidance.
2. Write the recursive form of `sumSeries()`. This function should use recursive function calls instead of a loop. Look at the *factorialByRecursion.py* program in this chapter for guidance.
3. Iteratively calculate the sum of the first n powers of 2 in a function named `sumPowersOf2()`. The powers of 2 are 2, 4, 8, 16, 32, and so on. In Python, these are calculated with `2 ** 1`, `2 ** 2`, `2 ** 3`, `2 ** 4`, `2 ** 5`, and so on, respectively. In JavaScript, these are calculated with `Math.pow(2, 1)`, `Math.pow(2, 2)`, and so on. For example, `sumPowersOf2(1)` returns 2, `sumPowersOf2(2)` returns 6 (that is, $2 + 4$), `sumPowersOf2(3)` returns 14 (that is, $2 + 4 + 8$), and so on.
4. Write the recursive form of `sumPowersOf2()`. This function should use recursive function calls instead of a loop.